



Contact

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Via Lusore 37, 30035,
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Education

2023 - present

MSc in Statistics
University of Padua, Italy

Relevant modules: Statistical Modeling, Statistical Inference, Data Analysis, Probability, Statistics for Partical Physics

2023

Exchange - Ludwig Maximilian
University of Munich, Germany

Relevant modules: Machine Learning, Regression for Correlated Data, Bayesian Statistics

2020- 2023

BSc in Statistics
University of Padua, Italy

Thesis title: "Clustering with Multilayer Neural Networks: A Feasibility Analysis on Astronomical Data"

Publications

Chiusaroli L., et al. (2026).
Outbreak of mucocutaneous Candida infections in newborns: clinical implications, investigation, and control measures. Antimicrobial Stewardship & Healthcare Epidemiology. doi:10.1017/ash.2025.10282

Languages

English - Advanced
Italian - Native

Marco Bolzonella

Master's student in Statistical Sciences with a strong research interest in mathematical statistics, likelihood-based inference, and non-standard asymptotic theory. Experienced in methodological research combining theoretical analysis with simulation studies, and in the practical implementation of statistical methods in R and Python.

Experience

September 2025 - February 2026

Nikhef, Amsterdam

Master's Thesis - ATLAS group

Conduct methodological research on likelihood-based statistical inference, with particular attention to non-standard settings and deviations from classical asymptotic results, in the context of SMEFT analyses within the ATLAS collaboration. Develop theoretical results for likelihood-based inference in non-standard settings and validate them through targeted simulation studies on the finite-sample behaviour of test statistics. Current work includes the preparation of one manuscript in statistical methodology and one physics-oriented manuscript related to the thesis project.

September 2024 - September 2025

Department of Cardio-Thoraco-Vascular Sciences
and Public Health, University of Padova

Research Assistant - Statistician

- Statistical analyses for research projects in public health
- Collaboration with multidisciplinary teams
- Writing scientific reports and academic papers

August 2023 - November 2023

University of Padua

Bachelor's Thesis

Conducted a methodological study on clustering methods based on multilayer neural networks, with a focus on model assessment and empirical validation. Analyzed performance and limitations of clustering approaches through simulation and application to astronomical datasets from the Gaia mission.

Skills

Technical & Analytical

- Likelihood-based statistical inference
- Asymptotic theory and non-regular inference
- Constrained models and boundary effects
- Monte Carlo simulation
- Statistical computing in R and Python

Reserch & Accademic

- Formulation of research questions and methodological problems
- Critical evaluation of statistical assumptions and models
- Scientific writing
- Presentation of research results
- Evidence synthesis & literature review

Reference

Alessandra Rosalba Brazzale

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Supervisor, University of Padua

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Lydia Brenner

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Supervisor, Nikhef

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